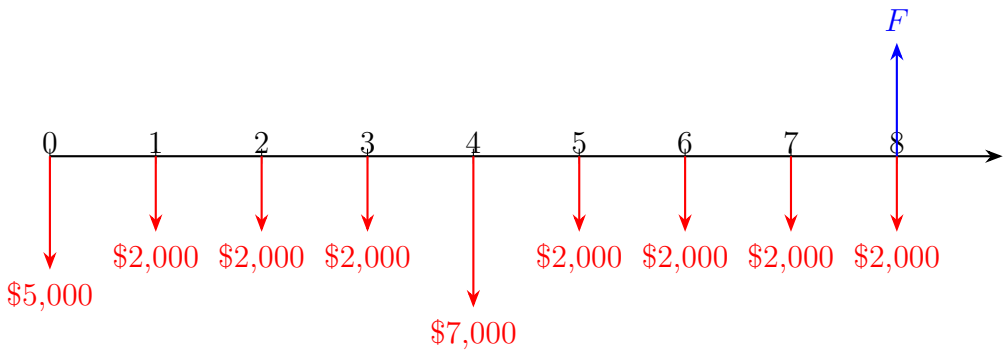
Homework Solutions
Chapter 2: How Time and Interest Affect Money

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Exercise 1: Single Amounts and Uniform Series

1. **Cash Flow Diagram:**



2. **Future Worth (F):**

$$F = 5000(F/P, 6\%, 8) + 2000(F/A, 6\%, 8) + 5000(F/P, 6\%, 4)$$
$$F = 5000(1.5938) + 2000(9.8975) + 5000(1.2625)$$
$$= 7969.24 + 19794.94 + 6312.38 = \$34,076.56$$

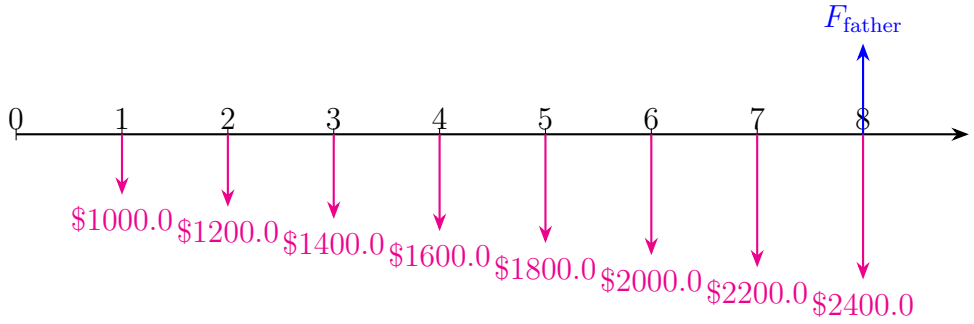
3. **Total Interest:** Total deposited $= \$5,000 + 8(\$2,000) + \$5,000 = \$26,000$
Interest $= \$34,076.56 - \$26,000 = \$8,076.56$

4. **Optional Challenge:** Since this replaces the single \$5,000 gift in Year 4, we first calculate the engineer's base savings without it, and then add the Future Worth of the father's arithmetic series.

   Engineer's Base Savings (without the Year 4 gift):

$$F_{\text{base}} = 5000(F/P, 6\%, 8) + 2000(F/A, 6\%, 8)$$
$$= 5000(1.5938) + 2000(9.8975) = 7969.24 + 19794.94 = \$27,764.18$$

   Father's Arithmetic Series ($A_1 = \$1,000$, $G = \$200$):



First, find the Present Worth of his gifts:

$$\begin{aligned} P_{\text{father}} &= 1000(P/A, 6\%, 8) + 200(P/G, 6\%, 8) \\ &= 1000(6.2098) + 200(19.8416) = 6209.80 + 3968.32 = \$10,178.12 \end{aligned}$$

Next, convert this to Future Worth at Year 8:

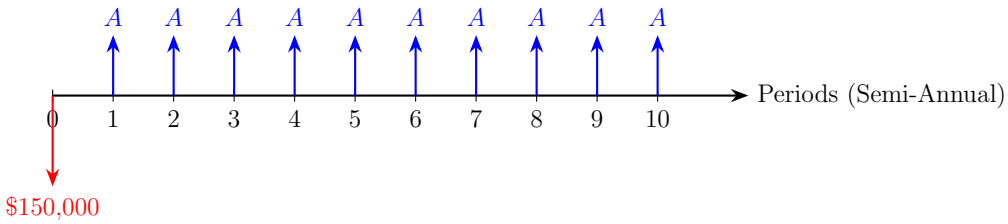
$$F_{\text{father}} = 10178.12(F/P, 6\%, 8) = 10178.12(1.5938) = \$16,221.89$$

Total Final Savings:

$$F_{\text{total}} = 27764.18 + 16221.89 = \$43,986.07$$

Exercise 2: Amortizing Loans and Capital Recovery

1. **CF Diagram (Bank's Perspective):**



2. **Semi-Annual Payment (A):** For 5 years with semi-annual payments, $n = 10$ periods. The interest rate per period is $8\%/2 = 4\%$.

$$A = 150000(A/P, 4\%, 10) = 150000(0.123291) = \$18,493.64$$

3. **Amortization Schedule (First 3 Years = 6 Periods):**

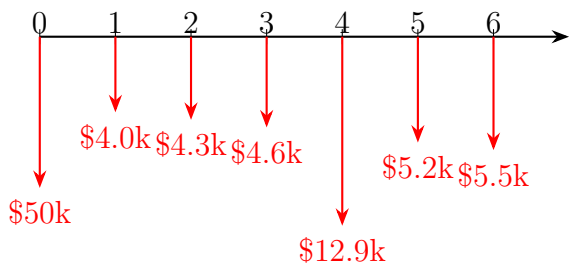
   - **Period 0:** Balance = \$150,000
   - **Period 1:** Interest $= 150,000 \times 0.04 = \$6,000$. Principal Paid $= 18,493.64 - 6,000 = \$12,493.64$. New Bal = \$137,506.36.
   - **Period 2:** Interest $= 137,506.36 \times 0.04 = \$5,500.25$. Principal Paid $= 18,493.64 - 5,500.25 = \$12,993.39$. New Bal = \$124,512.97.

- **Period 3:** Interest $= 124,512.97 \times 0.04 = \$4,980.52$. Principal Paid $= 18,493.64 - 4,980.52 = \$13,513.12$. New Bal = \$110,999.85.
- **Period 4:** Interest $= 110,999.85 \times 0.04 = \$4,439.99$. Principal Paid $= 18,493.64 - 4,439.99 = \$14,053.65$. New Bal = \$96,946.20.
- **Period 5:** Interest $= 96,946.20 \times 0.04 = \$3,877.85$. Principal Paid $= 18,493.64 - 3,877.85 = \$14,615.79$. New Bal = \$82,330.41.
- **Period 6:** Interest $= 82,330.41 \times 0.04 = \$3,293.22$. Principal Paid $= 18,493.64 - 3,293.22 = \$15,200.42$. New Bal = \$67,129.99.

Proportion of 3rd payment toward principal: $13,513.12/18,493.64 = 73.07\%$
Proportion of 3rd payment toward interest: $4,980.52/18,493.64 = 26.93\%$

Exercise 3: Composite Series and Arithmetic Gradient

1. **CF Diagram:**



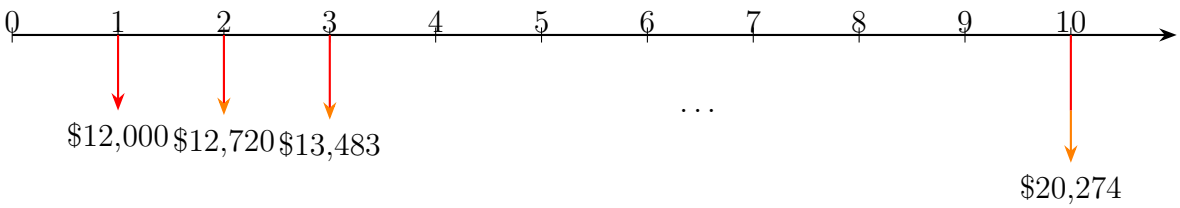
2. **Equivalent Present Worth (P) and Annual Worth (AW):**

$$
\begin{aligned}
P &= P_{\text{setup}} + P_{\text{soft}} + P_{\text{mech}} + P_{\text{overhaul}} \\
P_{\text{soft}} &= 2500(P/A, 7\%, 6) = 2500(4.7665) = \$11,916.35 \\
P_{\text{mech}} &= 1500(P/A, 7\%, 6) + 300(P/G, 7\%, 6) \\
&= 1500(4.7665) + 300(10.9784) = 7149.81 + 3293.52 = \$10,443.33 \\
P_{\text{overhaul}} &= 8000(P/F, 7\%, 4) = 8000(0.7629) = \$6,103.20 \\
\text{Total } P &= 50000 + 11916.35 + 10443.33 + 6103.20 = \$78,462.88 \\
AW &= P(A/P, 7\%, 6) = 78462.88(0.209796) = \$16,461.20
\end{aligned}
$$

3. **Inflow vs Outflow Interpretation:** They should be considered **outflows** (represented as downward arrows or negative values). Because the entire cash flow consists strictly of costs (installation, routine maintenance, overhauls), the equivalent P and AW mathematically represent the single equivalent upfront cost and the equivalent uniform annual cost required to sustain the robot.

Exercise 4: Geometric Gradient Problem Structuring

1. **CF Diagram:** $A_1 = \$12,000$, $g = 6\%$, $i = 12\%$, $n = 10$.



2. **Present Worth (P):** Since $i \neq g$:

$$\begin{aligned} P &= A_1 \left[\frac{\left(\frac{1+g}{1+i}\right)^n - 1}{g - i} \right] \\ &= 12000 \left[\frac{\left(\frac{1+0.06}{1+0.12}\right)^{10} - 1}{0.06 - 0.12} \right] \\ &= 12000(7.0549) \\ &= \$84,658.80 \end{aligned}$$

3. **Equivalent Flat Contract (A):**

$$A = P(A/P, 12\%, 10) = 84658.80(0.17698) = \$14,982.91$$

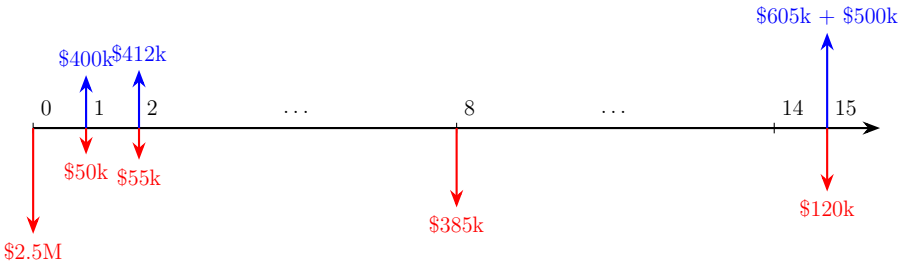
Exercise 5: Derivation of Factors

1. **Derivation of $(F/A, i, n)$:** The future worth is the sum of the compound amounts of each individual payment:

$$\begin{aligned} F &= A(1+i)^{n-1} + A(1+i)^{n-2} + A(1+i)^{n-3} + \cdots + A(1+i)^1 + A \\ &= A \sum_{j=0}^{n-1} (1+i)^j \\ &= A \frac{(1+i)^n - 1}{i} \end{aligned}$$

Exercise 7: Comprehensive Cash Flow Analysis

1. **CF Diagram:**



2. **Present Worth (P) and Viability:**

$$\begin{aligned} P_{\text{costs}} &= 2500000 + 50000(P/A, 10\%, 15) + 5000(P/G, 10\%, 15) + 300000(P/F, 10\%, 8) \\ &= 2500000 + 380305 + 200760 + 139950 = \$3,221,015 \\ P_{\text{revenues}} &= P_{\text{tolls}} + P_{\text{salvage}} \\ P_{\text{tolls}} &= 400000(P/A, g = 3\%, i = 10\%, n = 15) = 400000 \left[\frac{(1.03/1.10)^{15} - 1}{0.03 - 0.10} \right] \\ &= 400000(8.9576) = \$3,583,057 \\ P_{\text{salvage}} &= 500000(P/F, 10\%, 15) = 500000(0.2394) = \$119,696 \\ P_{\text{revenues}} &= \$3,702,753 \\ \text{Total } PW &= 3702753 - 3221016 = \$481,737 \end{aligned}$$

Is the handover cost fair? The total PW of the project is +$481,737. The $500,000 handover payment has a Present Worth of $119,696. Since the project's PW is highly positive even without the handover payment ($481,737 - $119,696 = $362,041), the project is already very profitable. Therefore, receiving an additional $500k handover is more than fair and highly favorable for the firm.

3. **Equivalent Uniform Annual Worth (AW):**

$$AW = PW(A/P, 10\%, 15) = 481737(0.13147) = \$63,336$$

Exercise 8: Present Worth of Contracts

1. **Total Present Worth (P):** The contract can be broken down into an immediate payment and two distinct uniform series. The second series must be shifted back to Year 0.

$$\begin{aligned} P &= 3,000,000 + 2,400,000(P/A, 8\%, 5) + 3,000,000(P/A, 8\%, 5)(P/F, 8\%, 5) \\ &= 3,000,000 + 2,400,000(3.9927) + 3,000,000(3.9927)(0.6806) \\ &= 3,000,000 + 9,582,504 + 3,000,000(2.71743) \\ &= 3,000,000 + 9,582,504 + 8,152,114 \\ &= \$20,734,618 \end{aligned}$$

Bo's "30 million" contract is actually worth $20.73 million at the time of signing due to the time value of money.

Exercise 9: Equipment Replacement and Viability

1. **Present Worth of Savings:** The savings represent a uniform series over 12 years.

$$P_{\text{savings}} = 35000(P/A, 9\%, 12) = 35000(7.1607) = \$250,624.50$$

2. **Conclusion:** The present worth of the operational savings ($250,624.50) exceeds the initial cost of the equipment ($215,000.00). The net present worth is positive ($35,624.50). *Yes, it is economically justifiable to buy the new equipment.*

Exercise 10: Equating Complex Arithmetic Gradients

1. **Present Worth of Withdrawals (Years 0-4):** The downward arrows represent a cash flow of \$1,500 at Year 0, followed by a decreasing arithmetic gradient from Year 1 to 4 with a base $A = \$1,200$ and $G = -\$300$.

$$\begin{aligned} PW_{\text{withdrawals}} &= 1500 + 1200(P/A, 6\%, 4) - 300(P/G, 6\%, 4) \\ &= 1500 + 1200(3.4651) - 300(4.9455) \\ &= 1500 + 4158.12 - 1483.65 \\ &= \$4,174.47 \end{aligned}$$

2. **Present Worth of Deposits (Years 6-10):** The upward arrows represent a standard arithmetic gradient starting at Year 6. Its "Year 0" equivalent is Year 5. Base $A = C$, Gradient $G = C$, $n = 5$.

$$\begin{aligned} P_5 &= C(P/A, 6\%, 5) + C(P/G, 6\%, 5) \\ &= C[4.2124 + 7.9345] \\ &= 12.1469C \end{aligned}$$

   Now, discount P_5 back to Year 0:

$$PW_{\text{deposits}} = P_5(P/F, 6\%, 5) = 12.1469C(0.7473) = 9.0773C$$

3. **Equating and Solving for C:**

$$\begin{aligned} PW_{\text{deposits}} &= PW_{\text{withdrawals}} \\ 9.0773C &= 4174.47 \\ C &= \frac{4174.47}{9.0773} = \$459.88 \end{aligned}$$

FE Exam Review Questions

1. **Answer: (d)**
   $F = G \times (P/G) \times (F/P)$. The factors mathematically chain together.

2. **Answer: (b)**
   The formula for (P/F) is $(1+i)^{-n}$. By exponent rules, $(1+i)^{-10} = (1+i)^{-4} \times (1+i)^{-6}$, so you multiply the factors.

3. **Answer: (b)**
   This is a decreasing arithmetic gradient with a base $A = \$800$ and $G = -\$100$.

$$A_{\text{total}} = 800 - 100(A/G, 4\%, 6) = 800 - 100(2.3936) = \$560.64$$

4. **Answer: (b)**
   Direct application of the Capital Recovery factor.

$$A = 100000(A/P, 10\%, 7) = 100000(0.20541) = \$20,541$$

5. **Answer: (a)**
   Direct application of the Sinking Fund factor.

$$A = 1500000(A/F, 10\%, 20) = 1500000(0.01746) = \$26,190$$

6. **Answer: (c)**
   This is a geometric gradient with $g = 3\%, i = 10\%, A_1 = \$5000, n = 20$. First, find the Present Worth:

$$P = 5000\left[\frac{1-(1.03/1.10)^{20}}{0.10-0.03}\right] = 5000\left[\frac{1-0.26829}{0.07}\right] = 5000(10.453) = \$52,265$$

   Then convert to Future Worth at Year 20:

$$F = 52265(F/P, 10\%, 20) = 52265(6.7275) = \$351,610$$

   This is closest to \$351,500.